Week 1: Vision Foundation and Data Encoding

Hybrid QML and Quantum String Algorithms for Document Extraction

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Deliverable as specified in the project brief: Technical summary of encoding efficiency versus classical normalisation.

1 Objective

Establish a quantum representation for document imagery. Two standard encodings were implemented and compared: FRQI (Flexible Representation of Quantum Images) and NEQR (Novel Enhanced Quantum Representation). The question to be answered is which is appropriate for a document-processing pipeline, and at what resource cost relative to simply normalising pixels classically.

2 The Two Encodings

FRQI stores each pixel's intensity as a rotation angle on a single shared colour qubit:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} \big(\cos\theta_i\,|0\rangle + \sin\theta_i\,|1\rangle\big) \otimes |i\rangle, \qquad \theta_i = \frac{\pi}{2} \cdot \frac{v_i}{255} \tag{1}$$

One qubit carries all colour information regardless of image size. The intensity range maps to $[0, \pi/2]$ rather than $[0, \pi]$ so that $\sin^2$ remains monotonic and a measured probability inverts to exactly one intensity.

NEQR stores intensity as a bit string on a colour register:

$$|I\rangle = \frac{1}{2^n} \sum_i |f(i)\rangle \otimes |i\rangle \tag{2}$$

An 8-bit greyscale image requires 8 colour qubits instead of 1, but reconstruction is exact.

3 Verification

A circuit that runs is not a circuit that is correct, so both encodings were inverted and checked against the original pixel values rather than assumed correct.

For FRQI, the colour qubit was measured conditioned on each position and the encoding inverted:

$$v_i = \frac{255 \cdot 2}{\pi} \arcsin \sqrt{P(\text{colour} = 1 \mid i)} \tag{3}$$

Table 1: Reconstruction accuracy on a 2×2 patch.

Encoding	Reconstruction	Maximum error
FRQI	statistical, from measurement frequencies	1–2 intensity levels of 255
NEQR	direct bit readout	**exactly 0**

The FRQI residual is sampling error and shrinks as $1/\sqrt{\text{shots}}$; the figure above is at 8,192 shots. NEQR's error is exactly zero because no estimation is involved.

4 Resource Comparison

Both circuits were transpiled to a common $\{u, \mathrm{CX}\}$ basis before counting. This matters for fairness: multi-controlled gates are single instructions at the abstract level but expand into very different numbers of two-qubit gates depending on their control count, so counting before transpilation would make a six-control gate appear as cheap as a CX.

Table 2: Measured resources across three patch sizes.

Patch	Pixels	FRQI qubits	NEQR qubits	FRQI CX	NEQR CX	Qubit ratio
2×2	4	3	10	32	114	3.33×
4×4	16	5	12	384	1,558	2.40×
8×8	64	7	14	3,584	10,998	2.00×

5 A Prediction, and Its Refutation

The initial version of this summary, written before the 4×4 and 8×8 measurements were taken, predicted that the qubit gap between the two encodings would **widen** as patch size grew.

Measurement refutes this. The ratio **narrows**: $3.33\times \to 2.40\times \to 2.00\times$.

The original reasoning conflated two distinct quantities. FRQI's colour register is fixed at 1 qubit and NEQR's at 8, so the colour gap is a *constant* 7 qubits at every patch size. Meanwhile both encodings share an identically growing position register of $\lceil \log_2(\text{pixels}) \rceil$. A constant gap divided by a growing shared base necessarily shrinks as a ratio.

The prediction was wrong and is recorded here rather than quietly corrected. Absolute CX counts do grow steeply for both encodings—NEQR reaches approximately 11,000 gates at 8×8—so NEQR remains far more expensive in absolute terms. It is specifically the *relative* qubit penalty that decays.

6 Comparison Against Classical Normalisation

The brief asked for encoding efficiency relative to classical normalisation. The comparison is stark and worth stating plainly.

Classical normalisation of a 2×2 patch is four floating-point divisions: approximately four operations, exact, with no measurement required. FRQI requires 3 qubits and 32 CX gates and returns an *approximate* value needing thousands of shots to estimate. NEQR requires 10 qubits and 114 CX gates for an exact value.

No efficiency argument favours quantum encoding at this scale. The encodings are of interest only as a route into subsequent quantum processing—if the subsequent processing provides an advantage, which is the question Weeks 2 and 4 address.

7 Consequence for the Pipeline

The pipeline ultimately uses *neither* encoding directly, and it is worth being explicit about why rather than leaving the reader to infer it.

The quanvolutional layer developed in Week 2 uses a simple R_Y angle encoding: one qubit per pixel of a 2×2 patch, rotated by $\pi v/255$. This is FRQI-like in spirit—intensity as angle—but has no position register, because a quanvolutional filter processes one patch at a time and never needs to address pixels by index.

FRQI and NEQR remain relevant as the resource baseline for *whole-image* quantum representation, which any approach operating on a full document at once would require. The measurement that NEQR costs roughly 11,000 CX gates for a single 8×8 patch is the quantitative reason this pipeline is patch-based rather than whole-image.

8 Deliverable Status

Complete. Both encodings implemented and verified by inversion; resource comparison measured at three patch sizes; the hypothesis about scaling tested and refuted.

Artifacts: `track_a_vision/encoding/encoding_scaling.py`, `notebooks/01_encoding_frqi_neqr.ipynb`, `benchmarking/encoding_scaling.json`